

• Sec-A ($5 \times 2 = 10$)

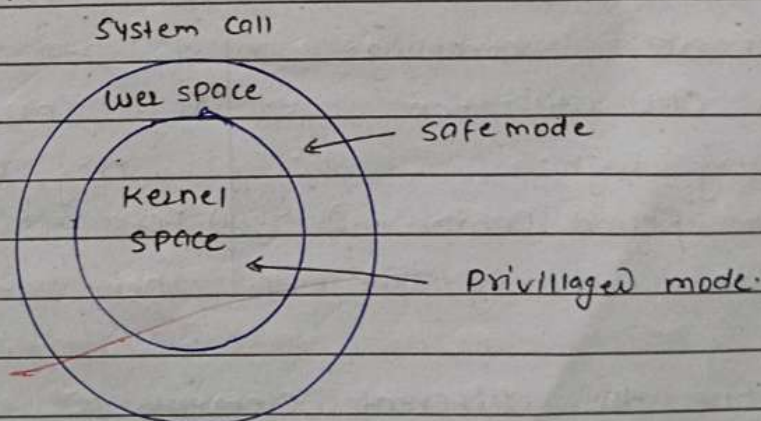
Q1. What is a process? Define different states of process.

Ans. A process is an instance of a running program. It has its own memory space, program counter and other resources. Processes can communicate with each other through shared memory, pipes and sockets.

- The different states of a process are:
- New: The process has just been created on this state.
- Ready: The process is ready to be executed but it is waiting for the CPU.
- Running: The process is currently being executed by the CPU.
- Waiting: The process is waiting for an event to occur, such as input from the user or completion of a disk I/O operation.
- Terminated: The process has finished executing and has been terminated by the OS.

Q2. What is system call?

Ans. A system call is a request made by a process to the operating system. System calls allow processes to access OS resources and services, such as reading and writing files, accessing the network and creating and deleting processes.



Q3. Define the following:

(0) Burst Time: Burst time is the amount of time that

A process needs to execute. It is also known as CPU time.

(b) Waiting Time: Waiting time is the amount of time that a process spends waiting in the ready queue to be executed.

(c) Turnaround Time: Turnaround time is the total amount of time that a process spends in the system, from the time it is created to the time it is terminated.

It is calculated as follows:

$$\text{Turnaround Time} = \text{Burst Time} + \text{Waiting Time}$$

(d) Arrival Time: Arrival time is the time at which a process enters the system.

Q4. Differentiate preemptive and non-preemptive scheduling algorithm.

<u>Ans.</u> Preemptive scheduling	Non-preemptive scheduling
(i) The resources are assigned to a process for a long time period.	Once resources are assigned to a process, they are held until it completes its burst period or changes to the waiting state.
(ii) It is flexible.	(i) It is rigid.
(iii) It has overheads associated with process scheduling.	(iii) It does not have overhead.
(iv) Its CPU utilization is very high.	(iv) Its CPU utilization is very low.
(v) ex: Round Robin and Shortest Remaining Time First.	(v) FCFS and SJF are examples of non-preemptive scheduling.

Q5. Define the different elements of a program.

Ans. Here are some of the essential elements of a program:

- Statements: A statement is an instruction that performs an action.

- **Functions:** A Function is a statement that returns a value.
- **Variables:** A Variable is a word defined in the Program that stores a value.
- **Operators:** An operator is an arithmetical symbol.
- **Objects:** An Object is a Program building block entity.
- **Method:** A Method is an action that an Object can perform.
- **Headers:** Headers are files that contain declaration of functions and variables that are used by the Program.

• **Section - B ($7 \times 3 = 21$)**

Q1. Demonstrate the concept of Context Switching with the help of block diagram.

Consider the set of 5 processes whose arrival time and burst time are given below -

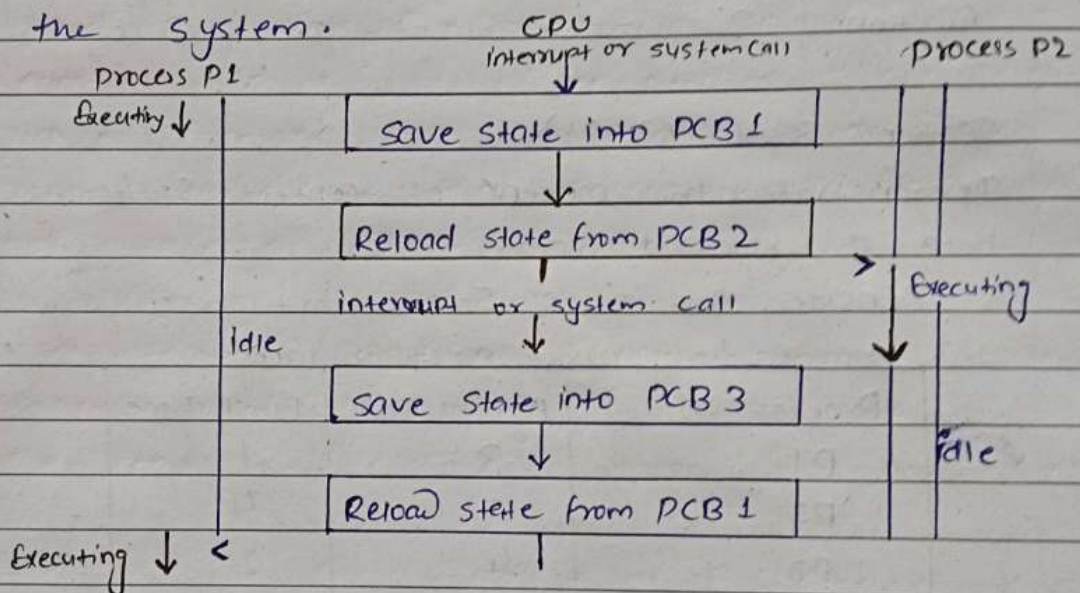
Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF preemptive, calculate the average waiting time and average turnaround time.

Ans.

Context switching: Context switching is a technique or method used by the operating system to switch a process from one state to another to execute its function using CPUs in the system. When switching perform in the system, it stores the old running process's status in the form of register and assigns the CPU to a new process to execute its

tasks. While a new process is running in the system, the previous process must wait in a ready queue. The execution of the old process starts at that point where another process stopped it. It defines the characteristics of a multitasking operating system in which multiple processes share the same CPU to perform multiple tasks without the need for additional processors in the system.



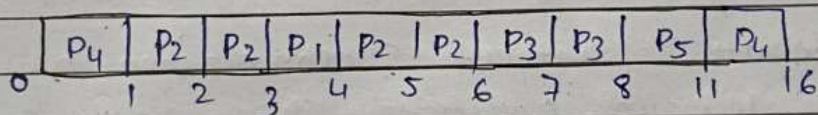
- Steps are taken when switching process P1 to process 2.
- (i) First, the context switching need to save the state of process P1 in the form of program.
- (ii) Now update PCB1 to process P1 and moves the process to the appropriate queue.
- (iii) After that, another process gets into the running state or we can select a new process from the ready state.
- (iv) Now, we have to update the PCB for the selected process P2.
- (v) If the CPU already executes process P2, we need to get the status of process P2 to resume its execution at the same time point where the system interrupt occurs.

Similar, Process P2 is switched off from the CPU so that the process P1 can resume execution.

#	Process Id	AT	BT	CT	TAT	WT	RT
	P1	3	10	4	1	0	0
	P2	1	4.5	6	5	1	0
	P3	4	20	8	4	2	2
	P4	0	65	16	16	10	0
	P5	2	30	11	9	6	6

- SJF Pre-emptive scheduling:

Gantt chart:



$$TAT = CT - AT$$

$$WT = TAT - BT$$

$$RT = \{ \text{CPU first time} - AT \}$$

- Average waiting Time = $\frac{0+1+2+10+6}{5} = \frac{19}{5} = 3.8 \text{ unit.}$

- Average TAT = $\frac{1+5+4+16+9}{5} = \frac{35}{5} = 7 \text{ unit.}$ Ans.

Q2. Define Concurrent Process. Explain the need of Process Synchronization in OS.

Ans. Concurrent Process refers to multiple processes that are executing independently and simultaneously in computer system. Each process has its own code, data & execution state and operates concurrently.

This can happen in a multitasking OS where

multiple Processes share a single CPU.

→ Process Synchronization is the Coordination of concurrent Processes to ensure that they access shared resources in a controlled and predictable manner. This is necessary to prevent race condition and other synchronization problems.

Need to Process Synchronization in OS are +

- (i) Resource sharing: it often need to share resources such as memory, files or hardware devices. It is essential to ensure that concurrent Processes do not Access share resources.
- (ii) Data Integrity: without Synchronization, multiple process accessing modifying shared data concurrently can lead to data inconsistency and corruption.
- (iii) ~~Deadlock~~ TO avoid Deadlock:- A deadlock is a situation where two or more Processes are waiting for each other to release a resource and none of the Processes can proceed.
- (iv) Optimize Resource Utilization:- it can help optimize resource utilization.

In the process synchronization there are three types of mechanism are included like - Semaphores, Monitors, and Condition variables.

Q3. Differentiate between: ~~Long~~^{Short} Term, Middle Term and Long Term schedulers.

Ans. Long Term scheduler: Long Term scheduler is also known as job scheduler. It is responsible for scheduling processes from the job pool and loading them into the ready queue in the main memory for execution. The long-term scheduler controls the degree of multiprogramming which is the number of processes in the ready queue at any given time.

- Medium-term scheduler: Medium-term scheduler is also known as swapping scheduler. It is responsible for swapping processes out of memory to secondary storage (such as a hard disk) and swapping them back in when needed. This is done to free up memory for other processes or to improve the performance of the system by keeping frequently used processes in memory.

- Short-term scheduler: Short-term scheduler is also known as CPU scheduler. It is responsible for selecting a process from the ready queue and assigning it to the CPU for execution. The short-term scheduler is typically invoked very frequently, as processes may be interrupted or preempted for various reasons.

In modern operating system, the long-term, medium-term, and short-term scheduler often work together to ensure that the system is running efficiently and that all processes are given a fair chance to execute.

• Difference :

	Long-Term Scheduler	Short-Term Scheduler	Medium-Term Scheduler
1.	It is a job scheduler.	It is a CPU scheduler.	It is a process swapping scheduler.
2.	It takes process from the job pool.	It takes process from the ready state.	It takes process from running/wait state.
3.	The speed of the short-term scheduler is very fast.	Speed of medium scheduler is in between the short-term and long-term scheduler.	The speed of a long-term scheduler is slow.
3.	Its speed is lesser than short-term scheduler.	Its speed is very fast among the both scheduler.	Its speed is in between long-term and short-term.
4.	It controls the degree of multiprogramming.	It has less control over the degree of multiprogramming.	It reduces the degree of multiprogramming.
5.	Absent or minimal in time sharing system.	Minimal in time sharing system.	Time sharing system use medium term scheduler.

• Part - C (11 x 3 = 33)

Q1. What is precedence graph? Explain the properties of Precedence graph with the help of an example.

Ans. Precedence Graph is a directed acyclic graph which is used to show the execution level of several processes in operating system. It consists of nodes and edges. Nodes represent the processes and the edges represent the flow of execution.

A precedence graph is acyclic, meaning that there are no cycles in the graph. This

means that there is no sequence of tasks that can be executed in such a way that a task depends on itself.

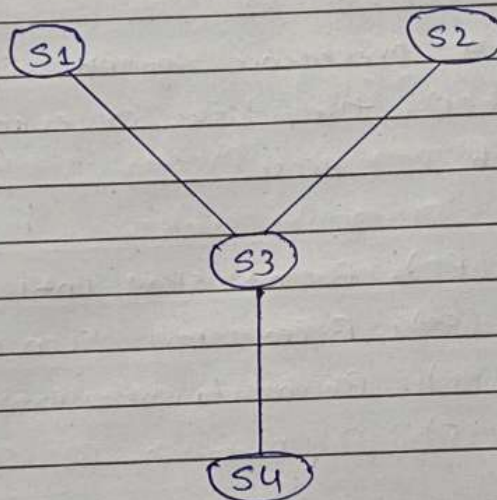
→ Consider the following code:

S1 : $a = x + y$;

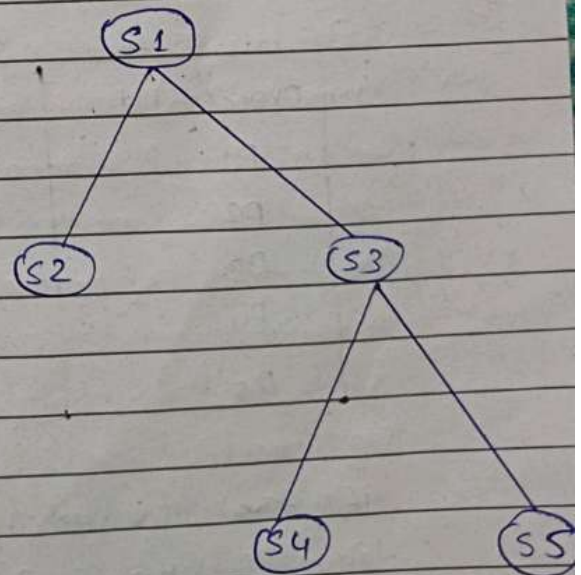
S2 : $b = z + 1$;

S3 : $c = a - b$;

S4 : $d = c + 1$;



- S1 per begin.
- S2, S3 per begin.
- S4, S5 per end.



• Properties of Precedence graph :-

- Directed : The edges in a precedence graph have a direction. This means that there is a clear order in which the tasks must be executed.
- Acyclic : There are no cycles in a precedence graph. This means that there is no sequence of tasks that can be executed in such a way that a task depends on itself.
- Complete : A precedence graph contains all of the dependencies betⁿ the tasks or events. This means that the graph is accu

• Finish-to-start (FS): B can't start until A finishes.

- Start-to-start (SS): B can start soon as A starts.
- Finish-to-finish (FF): B can't finish until A finishes.
- Start-to-finish (SF): B can finish as soon as A starts.

Q2. Consider the set of 5 processes whose arrival time and burst time are given below -

Process Id	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

If the CPU scheduling policy is Round Robin with time quantum = 2 unit, calculate the average waiting time and average turnaround time.

Ans	PID	AT	BT	CT	TAT	WT	RT
	P1	0	8:00	14	14	9	0
	P2	1	3:00	10	9	6	1
	P3	2	1:00	5	3	2	2
	P4	3	2:00	7	4	2	2
	P5	4	8:00	11	7	4	3

Time Quantum = 2 unit

Ready Queue = [P1 P2 P3 P4 P5 P2 P5 P1]

Running Queue = [P1 | P2 | P3 | P4 | P5 | P2 | P5 | P1]
 0 2 4 5 7 9 10 11 14

$$\text{Avg. WT} = \frac{9+6+2+2+4}{5} = \frac{23}{5} = 4.6 \text{ unit}$$

$$\text{Avg. TAT} = \frac{14+9+3+4+7}{5} = \frac{37}{5} = 7.4 \text{ unit}$$

Q3 Explain in detail: Process control block (PCB).

Ans. A process control block (PCB), also sometimes called a process descriptor, is a data structure used by computer operating systems to store all the information about a process. When a process is created, the operating system creates a corresponding process control block, which specifies and tracks the process state (i.e; new, ready, running, waiting or terminated). Since it is used to track process information, the PCB plays a key role in context switching.

The PCB is used by the operating system to manage and control the process. For example, when the operating system decides to switch to another process, it

Saves the current process's PCB to memory and loads the new process's PCB. This allows the operating system to quickly and efficiently switch between processes.

A process control block (PCB) contains information about the process, i.e.; registers, quantum, priority, etc. The process table is an array of PCBs, that means logically contains a PCB for all of the current processes in the system.

(PCB)

Stack Pointer	Process State
Program Counter	
Process Number	
CPU Register	
CPU scheduling Algorithm	
Memory Management	
Misc.	

- (i) **Stack Pointer**: it is a stack pointer that is required to be saved when the process is switched from one state to another to retain the current position of the process.
- (ii) **Process State**: it defines the current state of the process.
- (iii) **Program Counter**: it stores the counter, which contains the address of the next instruction that is to be executed for the process.



- (iv) **Process Number:** Every Process is assigned a unique id known as Process ID or PID which stores the Process identifier.
- (v) **CPU Register:** These are the CPU registers which include the accumulator, base, registers, and general-purpose registers.
- (vi) **CPU scheduling:** In this block which algorithms are used like, FCFS, Round robin, SJF, etc.
- (vii) **Memory Management:** This field contains the information about memory management system used by the operating system. This may include the list of files, opened for a process.

The PCB is a critical component of any operating system. it allows the operating system to manage and control multiple processes concurrently which is essential for modern computer systems.

• Advantages of PCB:

- Efficient Process management
- Resources management
- Process Synchronization.

• Disadvantages of PCB:

- overhead
- Complexity
- Miscellaneous accounting and status data.

NO.
18/10/23